

Reference-Free Relative and Absolute Anomaly Metrics for Generated Video Evaluation

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Abstract

We introduce reference-free anomaly metrics for generated video evaluation, combining relative and absolute measures of physical fidelity. Although tools like WorldGym or WorldEval enable robotic simulation via video generation, physical fidelity gaps often cannot reproduce real-world task success rates of VLA models. Unlike existing metrics, which require costly human voting (Elo) or unavailable ground-truth references (LPIPS), our metric utilizes DROID-SLAM and SEA-RAFT to quantify anomalies based on physical feasibility. Videos filtered with our relative metrics indicate improvement on task success rates by 12%, narrowing the simulation-to-reality gap. Furthermore, with our absolute metrics, visualization of when and where anomalies “jump” between frames is available.

1. Introduction

In robot learning, simulation has become a practical necessity due to the limited accessibility of real-world robotic systems. However, performance in simulation often fails to reliably transfer to real-world settings.

Recent VLA models including OpenVLA-OFT [3] show task success rate above 95% on LIBERO [5] benchmark. In contrast, more recent environments such as RoboCerebra [2] or VLABench [7] are sufficiently difficult that many existing policies struggle to achieve meaningful success rates. Fig. 1

World model-based approaches like WorldGym [6] or WorldEval [4] have shown promising alignment with real-world outcomes, suggesting a potential path toward more realistic evaluation. However, these approaches introduce a new challenge: the generated visual rollouts are often of limited quality, making it difficult to reliably assess task success.

While obvious artifacts (e.g., object distortion or morphing) can be easily identified, evaluating the overall quality of generated rollouts becomes challenging when multiple

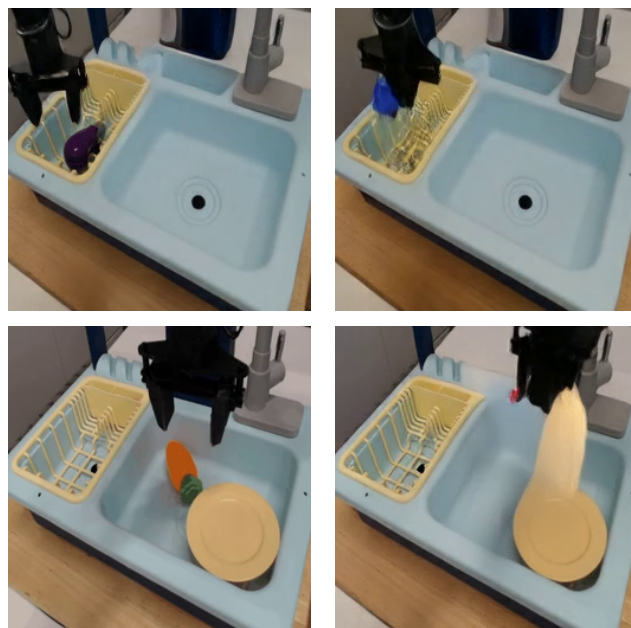


Figure 1. Artifacts in Generated Videos

imperfections are present. In such cases, it is unclear which samples are more suitable for evaluation.

To address this issue, we propose reference-free anomaly metrics to filter and detect physically implausible frames in generated videos.

2. Methods

Before deciding scoring methods, we generate 10 videos per task using WorldGym with random seed and OpenVLA for five out-of-distribution (OOD) tasks from 17 tasks used in both WorldGym and OpenVLA experiments. We select 5 tasks based on the availability of real-world experiment videos on the OpenVLA website, with one task representing each of the categories suggested in their study. The details of five tasks are summarized in Tab. 1.

Our approach is motivated by the WorldScore bench-

Category	Task	# Trials	WorldGym # Successes	Real-world # Successes
Visual gen	Put Eggplant into Pot (Easy Version)	10	7	10
Motion gen	Lift Eggplant	10	7.5	7.5
Physical gen	Put Carrot On Plate	10	4	8
Semantic gen	Stack Blue Cup on Pink Cup	10	6	4.5
Language Grounding	Put {Blue Cup, Pink Cup} on Plate	10	8.5	9.5
Mean Success Rate			66.0 $\pm$ 6.6%	79.0 $\pm$ 5.5%

Table 1. **Baseline Task Performance from Prior Work.** task success numbers are taken directly from WorldGym [6]. Average success rates $\pm$ StdErr are computed across 50 total rollouts, following OpenVLA calculated across 170 total rollouts for 17 tasks.

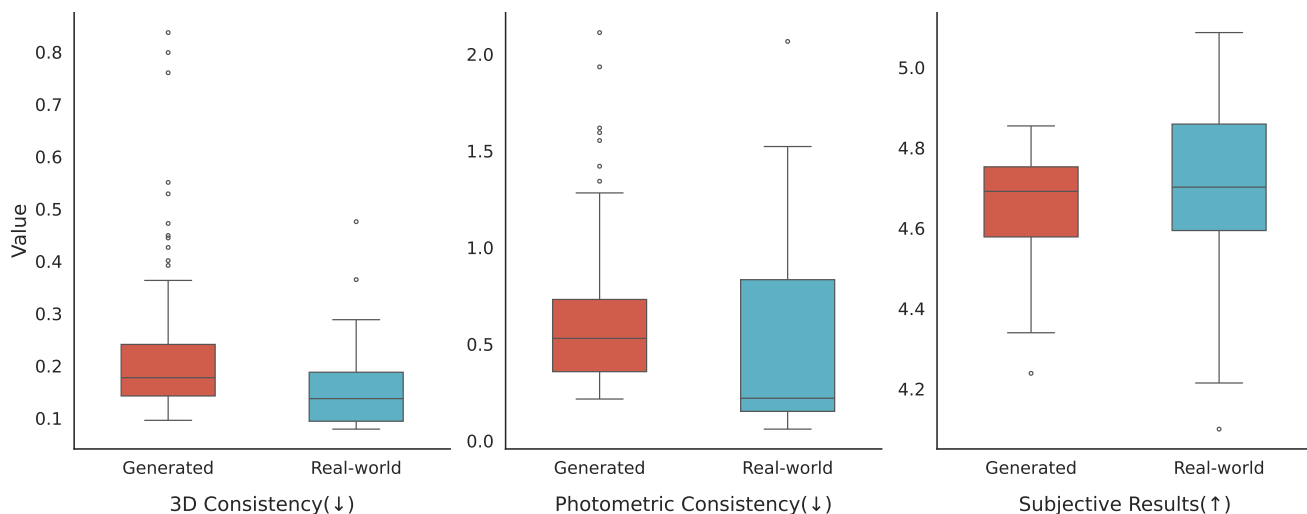


Figure 2. **Quality Scores in Real and Generated Videos.** 3D and photometric consistency exhibit distinctive gaps between real and generated distributions, while subjective quality is almost the same. Also, in generated videos, there are much more outliers in 3D and photometric consistency.

mark [1]. In particular, we focus on the quality metrics used in WorldScore. Among the four quality metrics in WorldScore, we exclude style consistency, as it is not directly related to physical consistency. We also exclude subjective quality, as the differences between real and generated videos are subtle. Consequently, we select 3D consistency and photometric consistency, since they indicate significant statistical gaps between real and generated videos as Fig. 2.

2.1. Relative Metrics

Our relative metrics are for filtering useful rollouts from various outputs from world models. To sort high and low anomaly rollouts, we generate 30 videos for each of the five tasks, resulting in a total of 150 videos. To ensure variability, each video was produced using a unique random seed, starting from 42 and incrementing for each subsequent generation.

For 3D consistency, we calculate the reprojection error across co-visible pixels between successive frames. In

terms of photometric consistency, we compute the Average End-Point Error(AEPE) with the optical flow. Both scores are computed using the publicly available source code from WorldScore [1].

To evaluate the informativeness of each consistency score for anomaly detection, we calculate the Shannon entropy $H(p) = -\sum_{i=1}^n p(x_i) \log p(x_i)$. With a total of 150 generated samples (30 samples per 5 distinct tasks) and 30 random real-world samples on OpenVLA websites, we put each score into $n = 30$ equal-width bins, where $p(x_i)$ denotes the frequency of scores falling into the i -th bin.

Our analysis reveals that the 3D consistency has a lower entropy compared to the photometric consistency. We interpret this lower entropy as a manifestation of higher informational certainty and representational stability. Consequently, we calculate weights with the stable softmax function, and assign each weight to the metrics in question. The

Metric	Property	Generated	Real-world
3D Consist	Entropy	2.16	1.73
	Weight	0.67	0.65
Photo Consist	Entropy	2.88	2.35
	Weight	0.33	0.35

Table 2. **Shannon Entropy and Weights of Real and Generated Videos.** The weights derived from generated and real-world datasets align, even with substantial disparity between two datasets.

weight is calculated as

$$w_k = \frac{\exp(-H_k)}{\sum_j \exp(-H_j)}$$

and specific numbers are described in Tab. 2.

Then, we get the relative anomaly score with the weighted sum, where $\hat{s}_k = (s_k - \mu_k)/\sigma_k$ denotes the z-score normalized score of metric k .

$$S_r = w_3 \cdot \hat{s}_3 + w_p \cdot \hat{s}_p$$

The distribution of relative anomaly scores reveals a clear shift, where generated videos tend to have higher scores and wider variance compared to real-world videos. This suggests that generated videos with low anomaly scores may better represent the real-world ability of VLA models.

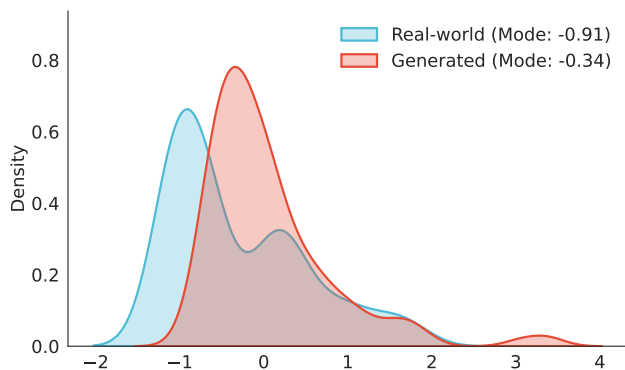


Figure 3. **Distribution of Relative Anomaly Scores.**

Following WorldGym [6], we use GPT-4o to evaluate each subset three times: videos with random, low, and high anomaly scores. The result is Tab. 3. The mean success rate is the highest with low anomaly videos, resulting in 12% increase over the WorldGym report. With random and high anomaly videos, performance showed little difference from the WorldGym report.

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Task	# Trials	Random	Low Anomaly	High Anomaly
Put Eggplant into Pot	10	6	9	9
Lift Eggplant	10	8	7	8
Put Carrot On Plate	10	5	7	3
Stack Blue Cup on Pink Cup	10	5	7	7
Put {Blue Cup, Pink Cup} on Plate	10	8	9	5
Mean Success Rate		64.0 ± 6.9%	78.0 ± 5.9%	64.0 ± 6.9%

Table 3. **Task Performance Evaluated After Filtering with Relative Anomaly Score.** Videos with low anomaly score were evaluated almost as the same success rate as the real world experiments.